

# ICRIC 2025

## International Conference on Research Innovations & Challenges (ICRIC 2025)

*Created & Powered by*

**Centre for Research & Training (CRT)**

National Foundation for Entrepreneurship Development (NFED)  
Coimbatore, Tamil Nadu, India

*In Association With*

**Technovate Educational & Consulting Services (TECS)**

Coimbatore, Tamil Nadu, India



**30<sup>th</sup> January - 1<sup>st</sup> February 2025**

**Edited By**

**KVM. Prof. Dr. R. Ganesan**

## **Appraisal of Groundwater Quality in and around Tummalapalle Uranium Mine at Kadapa District, Andhra Pradesh – A Chemometric Approach**

**Dr. Vangala Sunitha**

Professor

Department of Geology\*

&

**Ms. Parikisetti Deepthi**

Project Fellow

Department of Geology\*

&

**Dr. L. Dakshayani**

Associate Professor

Department of Genetics & Genomics\*

\*Yogi Vemana University

Kadapa, Andhra Pradesh, India

### **Abstract**

Drinking water and its purity is quite indispensable for a healthy living of mankind. Therefore, the purity of drinking water is often subjected to periodic assessments for alleviating contamination and taking precautionary measures for balancing the environment. The current study emphasizes on seasonal hydrogeochemistry, appraisal of groundwater quality for drinking purposes in the Tummalapalle uranium mining region, which is one of India's largest uranium reserves. The study included twenty groundwater samples from the study area. The physiochemical examination of groundwater is measured and compared to World Health organization (WHO) standards. Water samples are analysed for pH, Electrical Conductivity (EC), Total hardness (TH), Total dissolved solids (TDS), calcium ( $\text{Ca}^{2+}$ ), magnesium ( $\text{Mg}^{2+}$ ), sodium ( $\text{Na}^+$ ), potassium ( $\text{K}^+$ ), carbonates ( $\text{CO}_3^{2-}$ ), bicarbonates ( $\text{HCO}_3^-$ ), chlorides ( $\text{Cl}^-$ ), sulfates ( $\text{SO}_4^{2-}$ ) and nitrates ( $\text{NO}_3^-$ ). The results revealed that groundwater in the study area is neutral to slightly alkaline in nature. Moreover, the EC, total alkalinity, total hardness, magnesium, and sodium, nitrates exceed the WHO acceptable limit. The study utilized chemometric analysis and basic statistics, Gibbs ratio, and index of base exchange to understand the mechanism of controlling the groundwater chemistry in the study area. The present study indicated that the groundwater chemistry is mostly controlled by geogenic processes (weathering, dissolution, and ion exchange) and some extent of anthropogenic activities.